

A proof of Conjecture 00000007000

A permutation pair that preserves the box norm but changes the scattering spectrum

1 The conjecture

Conjecture 00000007000 reads:

Definition: Box norms and scattering spectra are two layers. *Conjecture:* There exist two functions with identical box norms but different scattering spectra, and the separation is realized by an explicit permutation pair with the same norms but different phase.

It is an existential statement. We prove it with two explicit real functions on the grid $\mathbb{Z}/4 \times \mathbb{Z}/4$.

2 Reading and definitions

Definition 1 (Box norm). For finite sets X, Y and $f : X \times Y \rightarrow \mathbb{R}$, the (Gowers) box norm is

$$\|f\|_{\square} = \left(\mathbb{E}_{x, x' \in X} \mathbb{E}_{y, y' \in Y} f(x, y) f(x, y') f(x', y) f(x', y') \right)^{1/4}.$$

The inner quantity equals $|X|^{-2} |Y|^{-2} \sum_{x, x'} (\sum_y f(x, y) f(x', y))^2 \geq 0$. For a function of two variables this is the box norm; we prove its invariance for every permutation pair.

Definition 2 (Permutation pair). A permutation pair is a pair (σ, τ) of a permutation σ of X and a permutation τ of Y . It acts by $f \mapsto f \circ (\sigma \times \tau)$, i.e. $(x, y) \mapsto f(\sigma x, \tau y)$.

Definition 3 (Scattering spectrum, phase). On $G = \mathbb{Z}/4 \times \mathbb{Z}/4$ the Fourier transform is $\hat{f}(\xi) = \sum_{(x, y) \in G} f(x, y) e^{-2\pi i(\xi_1 x + \xi_2 y)/4}$. The scattering spectrum is the diffraction intensity (structure factor) $I_f(\xi) = |\hat{f}(\xi)|^2$, and the phase at ξ is $\hat{f}(\xi)/|\hat{f}(\xi)|$.

In Lean the character is written i^k for $k \in \mathbb{Z}/4$ (using the representative $k \in \{0, 1, 2, 3\}$); the lemma `exp_eq` proves $i^k = e^{2\pi i k/4}$.

3 Results

Lemma 4 (Permutation pairs preserve the box norm). *For all f and every permutation pair, $\|f \circ (\sigma \times \tau)\|_{\square} = \|f\|_{\square}$.*

Proof. The sum over (x, x', y, y') is reindexed by the bijection $(x, x', y, y') \mapsto (\sigma x, \sigma x', \tau y, \tau y')$ of $X \times X \times Y \times Y$. $\square$

The explicit pair. Let $f(x, y) = 1$ if $x \in \{0, 1\}$ and $y = 0$, and $f = 0$ otherwise. Let $\sigma = (1\ 2)$ and $\tau = (0\ 1)$ (transpositions of $\mathbb{Z}/4$), and $g = f \circ (\sigma \times \tau)$. Then $g(x, y) = 1$ iff $x \in \{0, 2\}$ and $y = 1$.

Theorem 5 (Conjecture 00000007000). 1. $\sigma \neq \text{id}$, $\tau \neq \text{id}$, and $g = f \circ (\sigma \times \tau)$;
 2. *identical box norms*: $\|g\|_{\square} = \|f\|_{\square} > 0$;
 3. *different scattering spectra*: $I_f(1, 0) = 2$ while $I_g(1, 0) = 0$;
 4. *different phase*: $I_f(0, 1) = I_g(0, 1) = 4$, but the phase of $\hat{f}(0, 1)$ is 1 and that of $\hat{g}(0, 1)$ is $-i$.

Proof. (2) The equality is Lemma 4. The box-norm sum of f is $\sum_{x, x'} (\sum_y f(x, y) f(x', y))^2 = 4$, so $\|f\|_{\square} = (4/256)^{1/4} > 0$.

(3) $\hat{f}(1, 0) = 1 + e^{-2\pi i/4} = 1 - i$, so $I_f(1, 0) = 2$. $\hat{g}(1, 0) = 1 + e^{-2\pi i \cdot 2/4} = 1 - 1 = 0$.

(4) $\hat{f}(0, 1) = 2$ (both points have $y = 0$), while $\hat{g}(0, 1) = 2e^{-2\pi i/4} = -2i$. Both have modulus 2; the phases are 1 and $-i$. $\square$

Remark 6. A translation of $\mathbb{Z}/4$ is a permutation that changes only the phases of $\hat{f}$, never the intensities. The permutation $\sigma = (1\ 2)$ is not affine on $\mathbb{Z}/4$, which is why it can change the intensity at $(1, 0)$. For the displayed function f , which is supported on the single column $y = 0$, the transposition $\tau = (0\ 1)$ only moves that column to $y = 1$ and so multiplies $\hat{f}$ by a phase in the y direction; τ is not a translation, and for general functions it does change intensities. (This remark is explanatory only; it is not part of the Lean formalization.)

4 Lean formalization

The file `lean/Conjecture7000/Basic.lean` (Lean 4, Mathlib v4.33.1, namespace `Submission00000007000`) defines `boxSum`, `boxNorm` (with real power $1/4$), `permute`, `e4`, `fourier`, `intensity` (`Complex.normSq` of the Fourier coefficient) and `phase`. The main statements are `boxSum_eq_sq`, `boxSum_permute`, `boxNorm_permute` (Lemma 4), `exp_eq`, `g_eq`, `fourier_f_10`, `fourier_g_10`, `fourier_f_01`, `fourier_g_01`, `phase_f_01`, `phase_g_01`, `boxSum_f`, and

```
theorem conjecture7000 :
  exists (f g : G → ℝ) (s t : Equiv.Perm (ZMod 4)),
    s != 1 /\ t != 1 /\ g = permute s t f /\
    boxNorm g = boxNorm f /\ 0 < boxNorm f /\
    intensity f != intensity g /\ intensity f (1, 0) = 2 /\
    intensity g (1, 0) = 0 /\
    intensity f (0, 1) = intensity g (0, 1) /\
    phase f (0, 1) != phase g (0, 1)
```

(written here in ASCII; the Lean source uses Unicode symbols and the names σ, τ).

Axiom audit. `lake build` succeeds. `#print axioms` for `conjecture7000`, `boxNorm_permute` and `exp_eq` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry`, no `native_decide` and no added axiom.

5 Scope

“Box norm” is read as the Gowers box norm of a real function of two variables; “scattering spectrum” as the diffraction intensity $|\hat{f}|^2$ on the finite grid; “permutation pair” as a pair of coordinate

permutations; “phase” as the unit phase of a Fourier coefficient. Under this reading both the intensity and the phase change. We do not treat the wavelet scattering transform (Mallat) or quantum-mechanical scattering matrices.